Universidad del Rosario | Escuela de Ciencias e Ingeniería | DEPE

INTRODUCCIÓN A R Y RSTUDIO

SÁBADO
9 DE AGOSTO

10:00 AM A 12:00 PM

En esta sesión de dos horas, exploraremos los conceptos esenciales de R y RStudio, dos herramientas poderosas para el análisis de datos y la visualización.

Taller Knuth

MODALIDAD ACCESO REMOTO

NO SE REQUIERE EXPERIENCIA PREVIA EN PROGRAMACIÓN

Temas

- ¿Qué es R?
- R y RStudio
- Instalación
- Interfaz de RStudio
- Programación
  - Componentes
  - Estructuras de control
  - Funciones

¿Qué es R?

- Dialecto de S. John Chambers. Bell Labs AT&T 1976

> "[W]e wanted users to be able to begin in an interactive environment, where they did not consciously think of themselves as programming. Then as their needs became clearer and their sophistication increased, they should be able to slide gradually into programming, when the language and system aspects would become more important."

- R. Ross Ihaka y Robert Gentleman. Universidad de Auckland. Nueva Zelanda. 1993
- GNU General Public License. 1995
- R Versión 1.0.0 Released. 2000

¿Qué es R?

Corre en casi todas las plataformas de computación y Sos

Open Source

Releases muy frecuentes

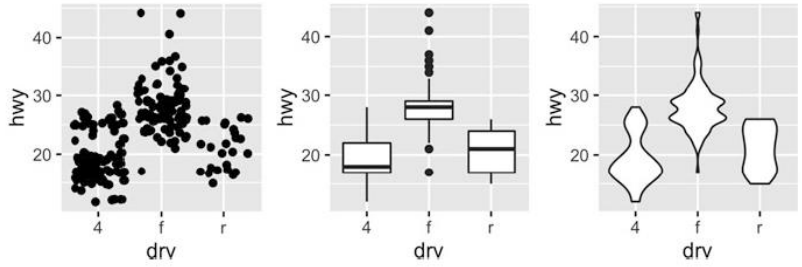
Capacidades gráficas sofisticadas

Programación interactiva

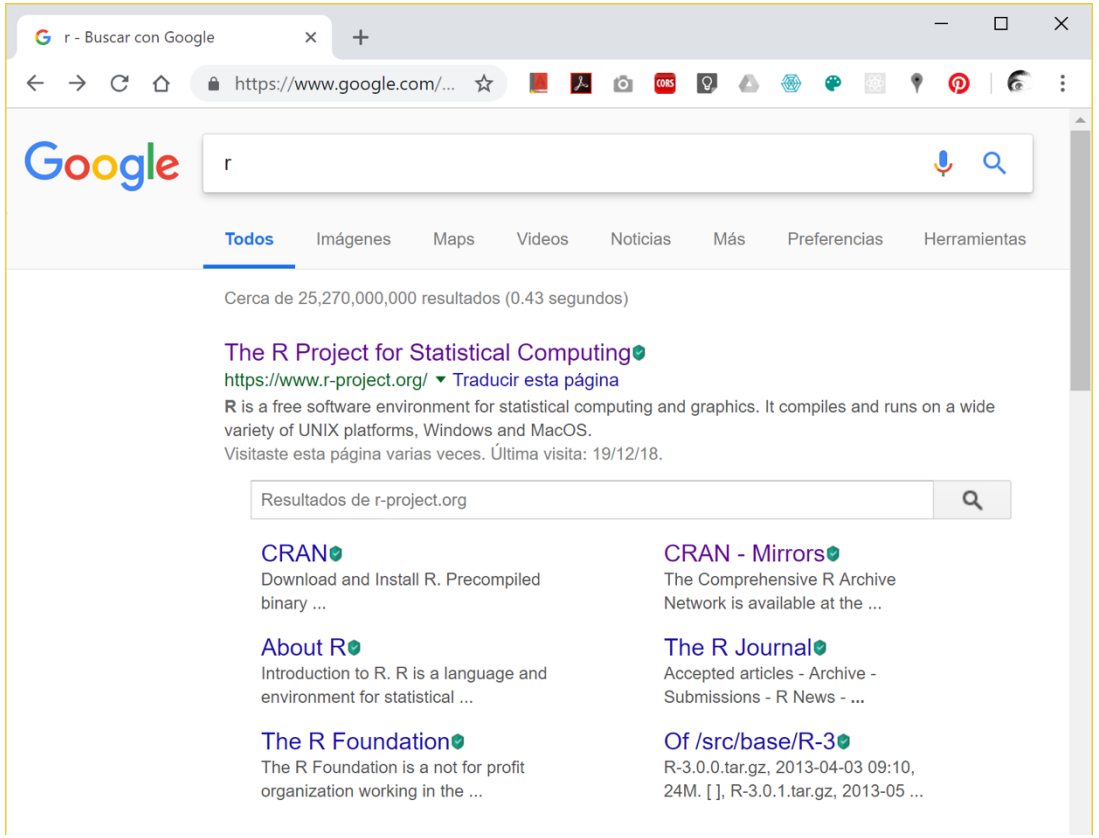
Poderoso lenguaje de programación

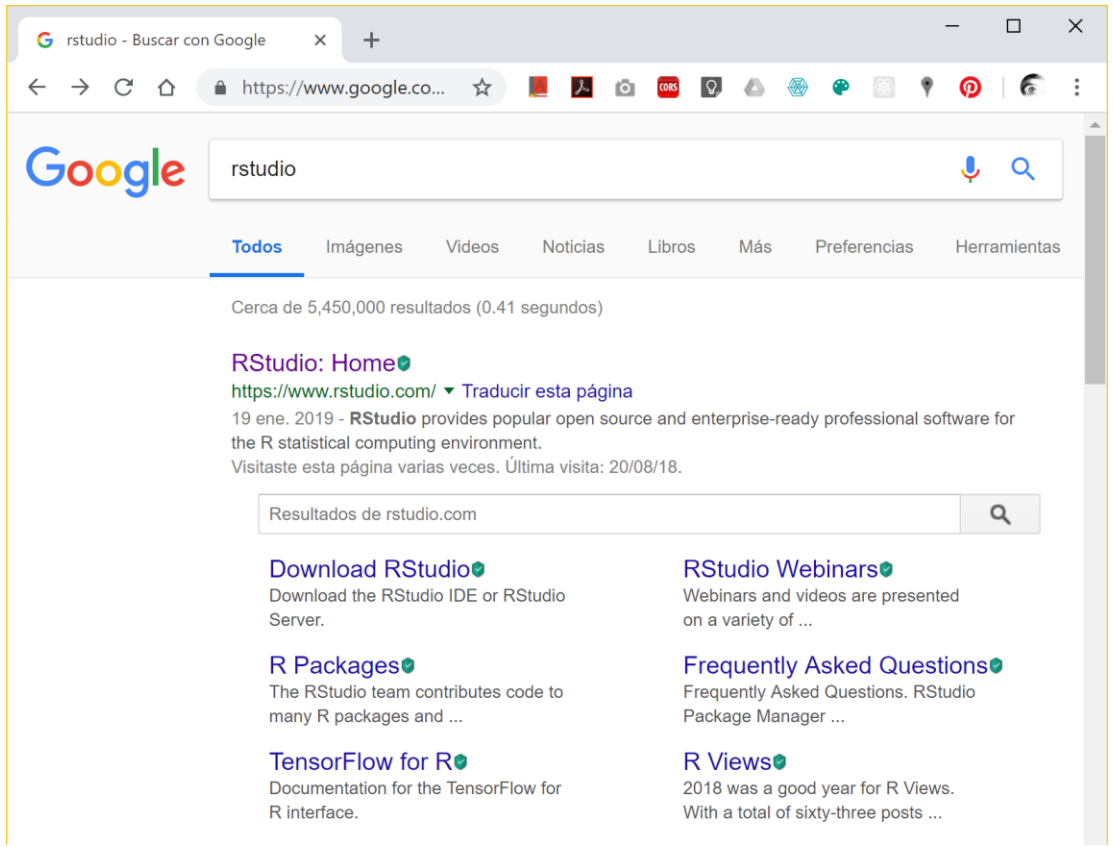
Stack Overflow

```
ggplot(mpg, aes(drv, hwy)) + geom_jitter()
ggplot(mpg, aes(drv, hwy)) + geom_boxplot()
ggplot(mpg, aes(drv, hwy)) + geom_violin()
```

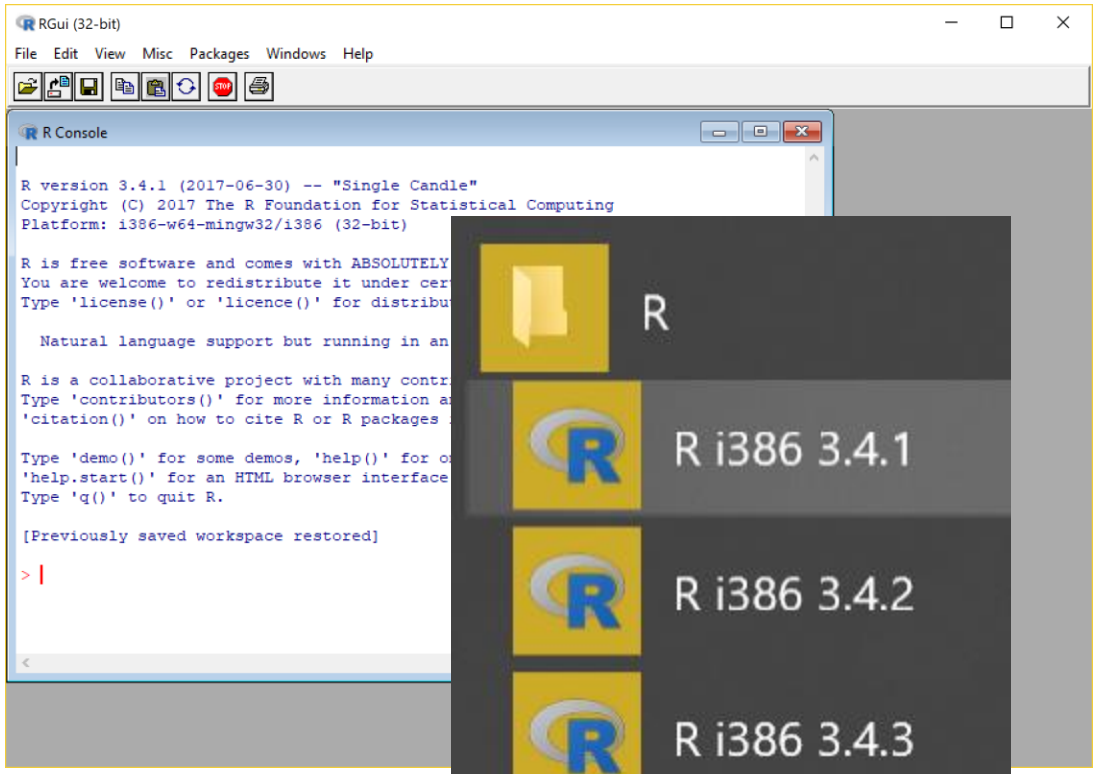


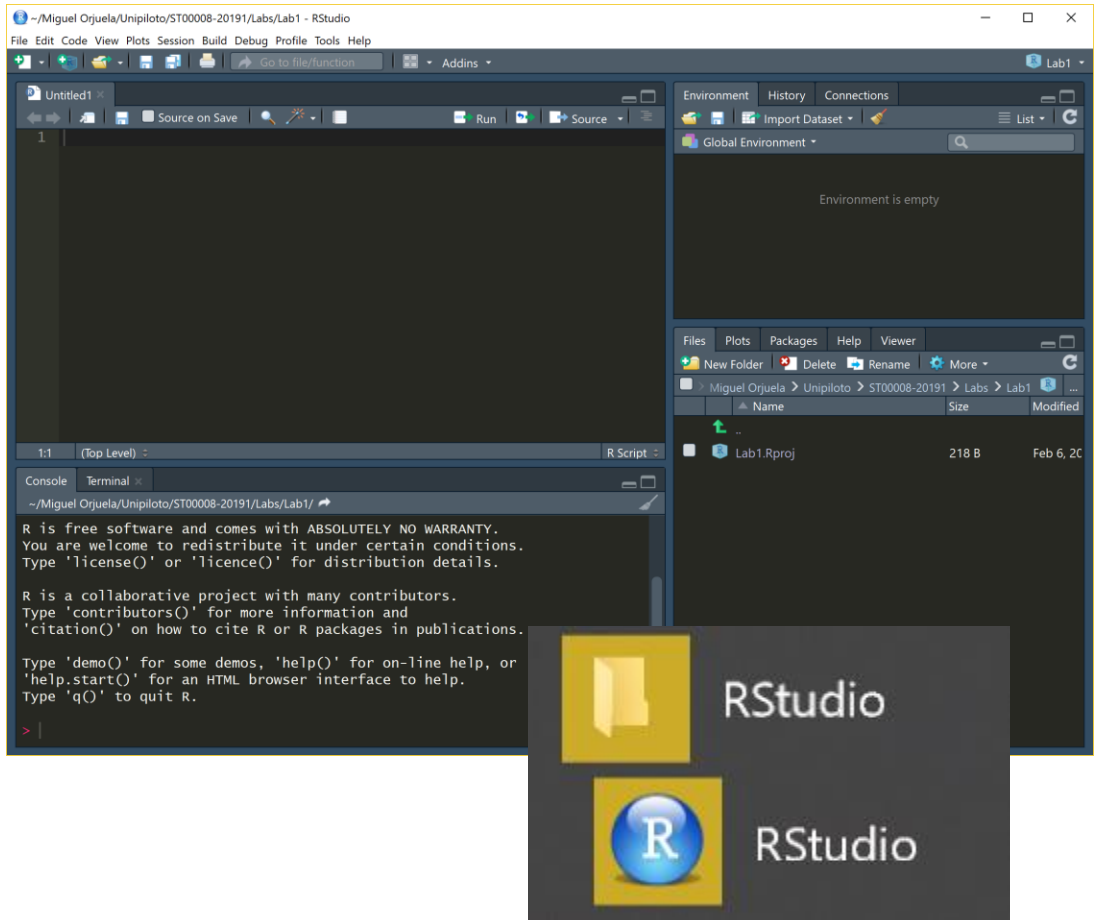
R y RStudio


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The R Project for Statistical Computing
https://www.r-project.org/ ▾ Traducir esta página
R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS.
Visitaste esta página varias veces. Última visita: 19/12/18.
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The R Foundation
The R Foundation is a not for profit organization working in the ...
Of /src/base/R-3
R-3.0.0.tar.gz, 2013-04-03 09:10, 24M. [ ], R-3.0.1.tar.gz, 2013-05 ...

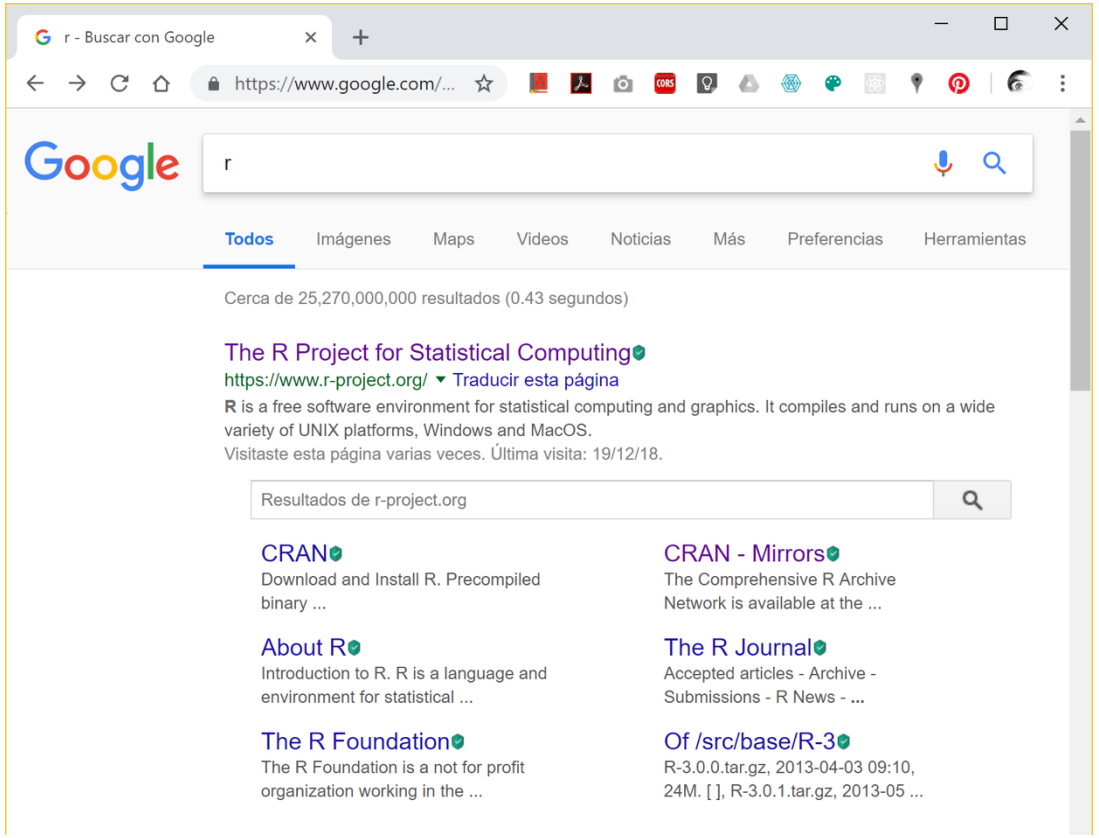

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RStudio: Home
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19 ene. 2019 - RStudio provides popular open source and enterprise-ready professional software for the R statistical computing environment.
Visitaste esta página varias veces. Última visita: 20/08/18.
Resultados de rstudio.com
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R Packages
The RStudio team contributes code to many R packages and ...
Frequently Asked Questions
Frequently Asked Questions. RStudio Package Manager ...
TensorFlow for R
Documentation for the TensorFlow for R interface.
R Views
2018 was a good year for R Views. With a total of sixty-three posts ...

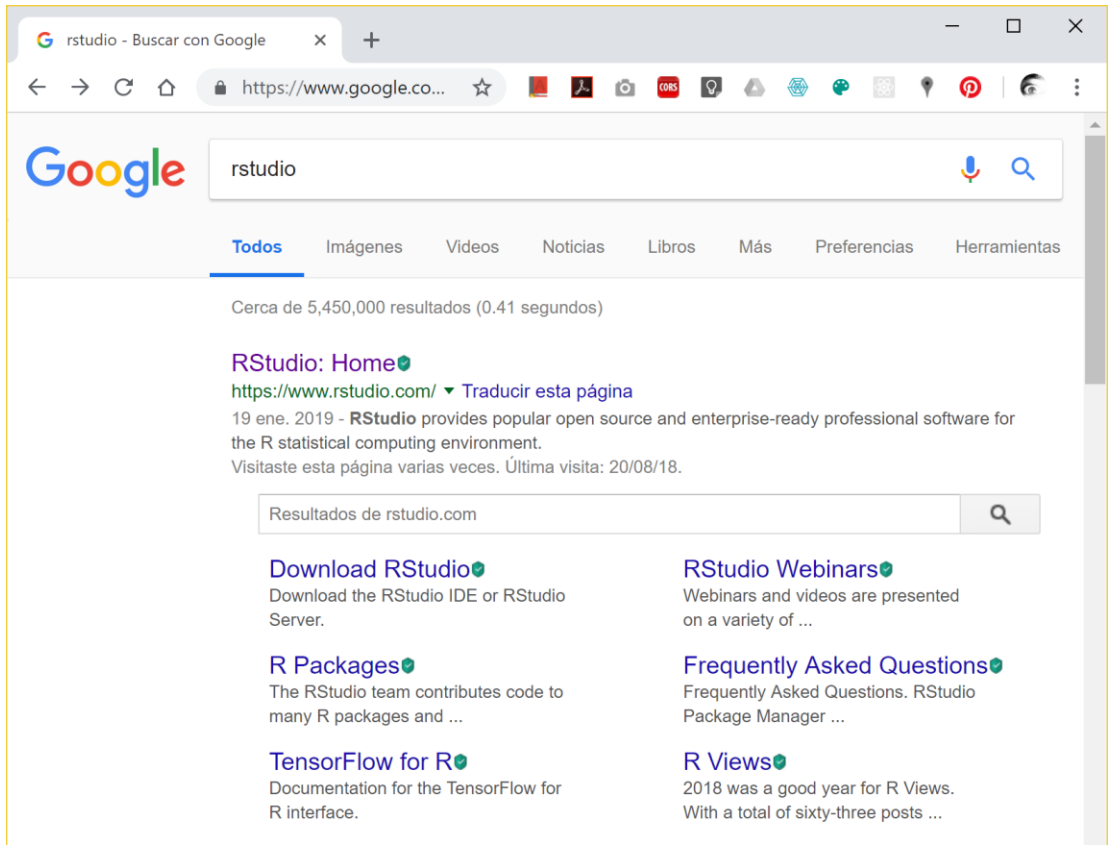
R y RStudio


RGui (32-bit)
File Edit View Misc Packages Windows Help
R Console
R version 3.4.1 (2017-06-30) -- "Single Candle"
Copyright (C) 2017 The R Foundation for Statistical Computing
Platform: i386-w64-mingw32/i386 (32-bit)
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You are welcome to redistribute it under cer
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Natural language support but running in an
R is a collaborative project with many contr
Type 'contributors()' for more information a
'citation()' on how to cite R or R packages
Type 'demo()' for some demos, 'help()' for o
'help.start()' for an HTML browser interface
Type 'q()' to quit R.
[Previously saved workspace restored]
R
R i386 3.4.1
R i386 3.4.2
R i386 3.4.3
R i386 3.5.1

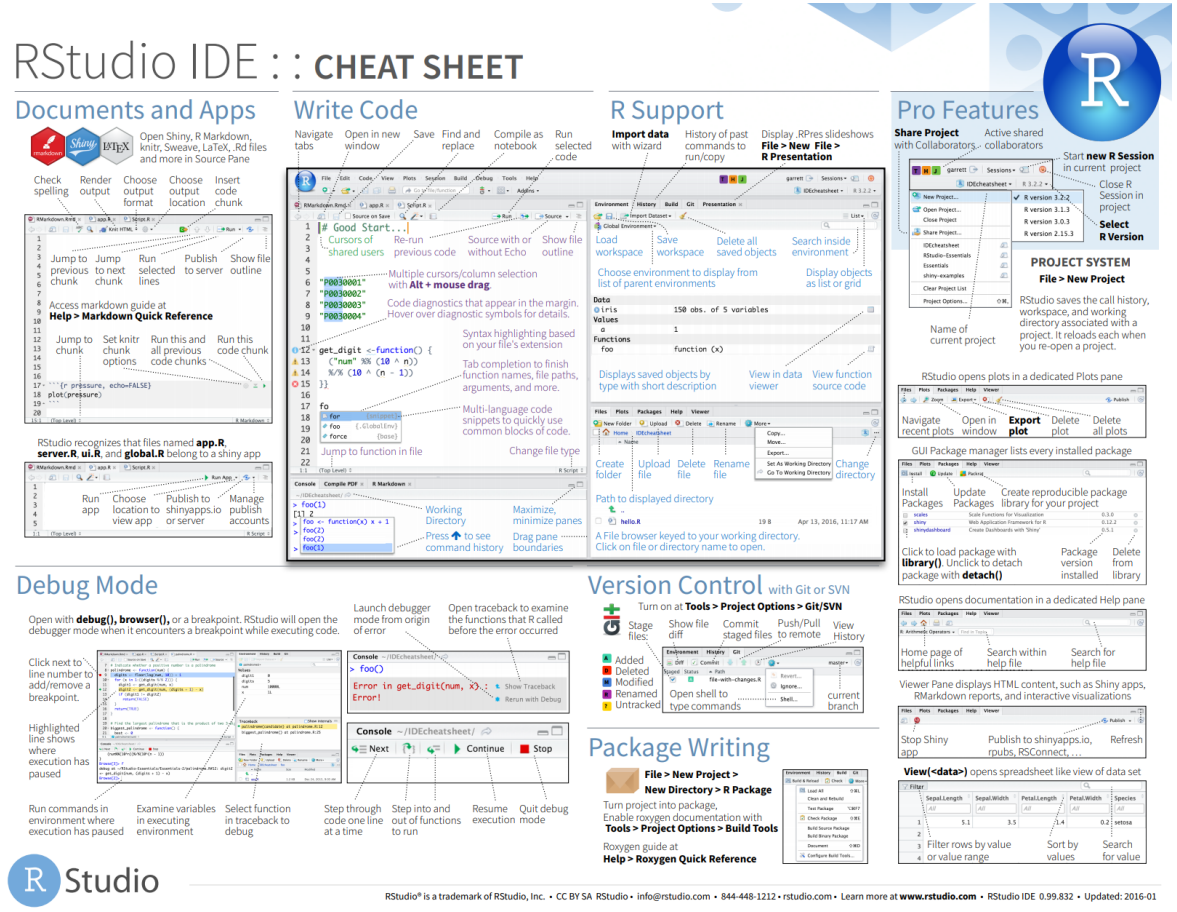

~/Miguel Orjuela/Unipiloto/ST00008-20191/Labs/Lab1 - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1
Source on Save
Run
Source
Environment History Connections
Import Dataset
Global Environment
Environment is empty
Files Plots Packages Help Viewer
New Folder Delete Rename More
Name Size Modified
Lab1.Rproj 218 B Feb 6, 20
Console Terminal
~/Miguel Orjuela/Unipiloto/ST00008-20191/Labs/Lab1/
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R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
RStudio
RStudio

Instalación

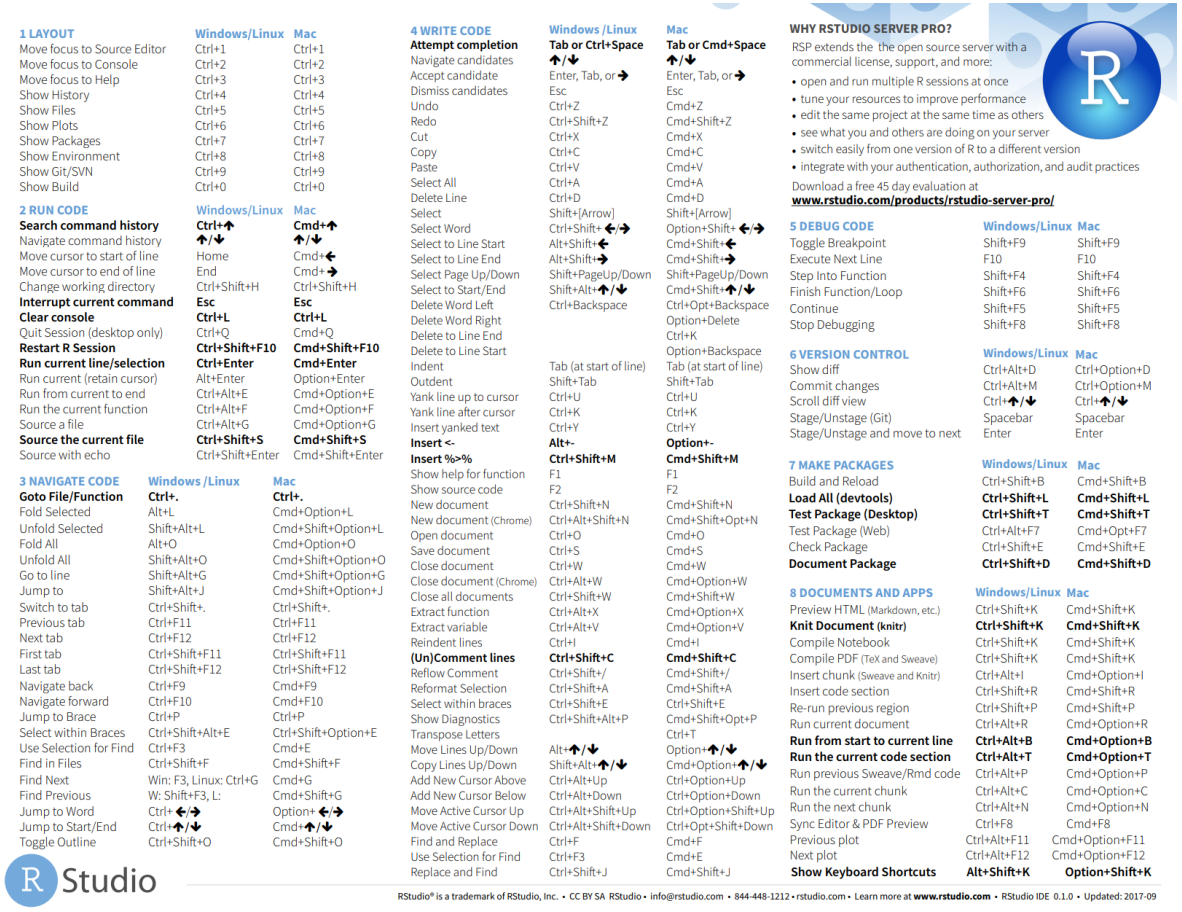




Interfaz de RStudio



Interfaz de RStudio



1 LAYOUT

	Windows/Linux	Mac
Move focus to Source Editor	Ctrl+1	Ctrl+1
Move focus to Console	Ctrl+2	Ctrl+2
Move focus to Help	Ctrl+3	Ctrl+3
Show History	Ctrl+4	Ctrl+4
Show Files	Ctrl+5	Ctrl+5
Show Plots	Ctrl+6	Ctrl+6
Show Packages	Ctrl+7	Ctrl+7
Show Environment	Ctrl+8	Ctrl+8
Show Git/SVN	Ctrl+9	Ctrl+9
Show Build	Ctrl+0	Ctrl+0

2 RUN CODE

	Windows/Linux	Mac
Search command history	**Ctrl+↑**	**Cmd+↑**
Navigate command history	**↑/↓**	**↑/↓**
Move cursor to start of line	Home	Cmd+←
Move cursor to end of line	End	Cmd+→
Change working directory	Ctrl+Shift+H	Ctrl+Shift+H
Interrupt current command	**Esc**	**Esc**
Clear console	**Ctrl+L**	**Ctrl+L**
Quit Session (desktop only)	Ctrl+Q	Cmd+Q
Restart R Session	**Ctrl+Shift+F10**	**Cmd+Shift+F10**
Run current line/selection	**Ctrl+Enter**	**Cmd+Enter**
Run current (retain cursor)	Alt+Enter	Option+Enter
Run from current to end	Ctrl+Alt+E	Cmd+Option+E
Run the current function	Ctrl+Alt+F	Cmd+Option+F
Source a file	Ctrl+Alt+G	Cmd+Option+G
Source the current file	**Ctrl+Shift+S**	**Cmd+Shift+S**
Source with echo	Ctrl+Shift+Enter	Cmd+Shift+Enter

3 NAVIGATE CODE

	Windows /Linux	Mac
Goto File/Function	**Ctrl+.**	**Ctrl+.**
Fold Selected	Alt+L	Cmd+Option+L
Unfold Selected	Shift+Alt+L	Cmd+Shift+Option+L
Fold All	Alt+O	Cmd+Option+O
Unfold All	Shift+Alt+O	Cmd+Shift+Option+O
Go to line	Shift+Alt+G	Cmd+Shift+Option+G
Jump to	Shift+Alt+J	Cmd+Shift+Option+J
Switch to tab	Ctrl+Shift+.	Ctrl+Shift+.
Previous tab	Ctrl+F11	Ctrl+F11
Next tab	Ctrl+F12	Ctrl+F12
First tab	Ctrl+Shift+F11	Ctrl+Shift+F11
Last tab	Ctrl+Shift+F12	Ctrl+Shift+F12
Navigate back	Ctrl+F9	Cmd+F9
Navigate forward	Ctrl+F10	Cmd+F10
Jump to Brace	Ctrl+P	Ctrl+P
Select within Braces	Ctrl+Shift+Alt+E	Ctrl+Shift+Option+E
Use Selection for Find	Ctrl+F3	Cmd+E
Find in Files	Ctrl+Shift+F	Cmd+Shift+F
Find Next	Win: F3, Linux: Ctrl+G	Cmd+G
Find Previous	W: Shift+F3, L:	Cmd+Shift+G
Jump to Word	Ctrl+ ←/→	Option+ ←/→
Jump to Start/End	Ctrl+↑/↓	Cmd+↑/↓
Toggle Outline	Ctrl+Shift+O	Cmd+Shift+O

4 WRITE CODE

	Windows /Linux	Mac
Attempt completion	**Tab or Ctrl+Space**	**Tab or Cmd+Space**
Navigate candidates	**↑/↓**	**↑/↓**
Accept candidate	Enter, Tab, or →	Enter, Tab, or →
Dismiss candidates	Esc	Esc
Undo	Ctrl+Z	Cmd+Z
Redo	Ctrl+Shift+Z	Cmd+Shift+Z
Cut	Ctrl+X	Cmd+X
Copy	Ctrl+C	Cmd+C
Paste	Ctrl+V	Cmd+V
Select All	Ctrl+A	Cmd+A
Delete Line	Ctrl+D	Cmd+D
Select	Shift+[Arrow]	Shift+[Arrow]
Select Word	Ctrl+Shift+ ←/→	Option+Shift+ ←/→
Select to Line Start	Alt+Shift+←	Cmd+Shift+←
Select to Line End	Alt+Shift+→	Cmd+Shift+→
Select Page Up/Down	Shift+PageUp/Down	Shift+PageUp/Down
Select to Start/End	Shift+Alt+↑/↓	Cmd+Shift+↑/↓
Delete Word Left	Ctrl+Backspace	Ctrl+Opt+Backspace
Delete Word Right		Option+Delete
Delete to Line End		Ctrl+K
Delete to Line Start		Option+Backspace
Indent	Tab (at start of line)	Tab (at start of line)
Outdent	Shift+Tab	Shift+Tab
Yank line up to cursor	Ctrl+U	Ctrl+U
Yank line after cursor	Ctrl+K	Ctrl+K
Insert yanked text	Ctrl+Y	Ctrl+Y
Insert <-	**Alt+-**	**Option+-**
Insert %>%	**Ctrl+Shift+M**	**Cmd+Shift+M**
Show help for function	F1	F1
Show source code	F2	F2
New document	Ctrl+Shift+N	Cmd+Shift+N
New document (Chrome)	Ctrl+Alt+Shift+N	Cmd+Shift+Opt+N
Open document	Ctrl+O	Cmd+O
Save document	Ctrl+S	Cmd+S
Close document	Ctrl+W	Cmd+W
Close document (Chrome)	Ctrl+Alt+W	Cmd+Option+W
Close all documents	Ctrl+Shift+W	Cmd+Shift+W
Extract function	Ctrl+Alt+X	Cmd+Option+X
Extract variable	Ctrl+Alt+V	Cmd+Option+V
Reindent lines	Ctrl+I	Cmd+I
(Un)Comment lines	**Ctrl+Shift+C**	**Cmd+Shift+C**
Reflow Comment	Ctrl+Shift+/	Cmd+Shift+/
Reformat Selection	Ctrl+Shift+A	Cmd+Shift+A
Select within braces	Ctrl+Shift+E	Ctrl+Shift+E
Show Diagnostics	Ctrl+Shift+Alt+P	Cmd+Shift+Opt+P
Transpose Letters		Ctrl+T
Move Lines Up/Down	Alt+↑/↓	Option+↑/↓
Copy Lines Up/Down	Shift+Alt+↑/↓	Cmd+Option+↑/↓
Add New Cursor Above	Ctrl+Alt+Up	Ctrl+Option+Up
Add New Cursor Below	Ctrl+Alt+Down	Ctrl+Option+Down
Move Active Cursor Up	Ctrl+Alt+Shift+Up	Ctrl+Option+Shift+Up
Move Active Cursor Down	Ctrl+Alt+Shift+Down	Ctrl+Opt+Shift+Down
Find and Replace	Ctrl+F	Cmd+F
Use Selection for Find	Ctrl+F3	Cmd+E
Replace and Find	Ctrl+Shift+J	Cmd+Shift+J

WHY RSTUDIO SERVER PRO?

RSP extends the the open source server with a commercial license, support, and more:

- open and run multiple R sessions at once
- tune your resources to improve performance
- edit the same project at the same time as others
- see what you and others are doing on your server
- switch easily from one version of R to a different version
- integrate with your authentication, authorization, and audit practices

Download a free 45 day evaluation at
www.rstudio.com/products/rstudio-server-pro/

5 DEBUG CODE

	Windows/Linux	Mac
Toggle Breakpoint	Shift+F9	Shift+F9
Execute Next Line	F10	F10
Step Into Function	Shift+F4	Shift+F4
Finish Function/Loop	Shift+F6	Shift+F6
Continue	Shift+F5	Shift+F5
Stop Debugging	Shift+F8	Shift+F8

6 VERSION CONTROL

	Windows/Linux	Mac
Show diff	Ctrl+Alt+D	Ctrl+Option+D
Commit changes	Ctrl+Alt+M	Ctrl+Option+M
Scroll diff view	Ctrl+↑/↓	Ctrl+↑/↓
Stage/Unstage (Git)	Spacebar	Spacebar
Stage/Unstage and move to next	Enter	Enter

7 MAKE PACKAGES

	Windows/Linux	Mac
Build and Reload	Ctrl+Shift+B	Cmd+Shift+B
Load All (devtools)	**Ctrl+Shift+L**	**Cmd+Shift+L**
Test Package (Desktop)	**Ctrl+Shift+T**	**Cmd+Shift+T**
Test Package (Web)	Ctrl+Alt+F7	Cmd+Opt+F7
Check Package	Ctrl+Shift+E	Cmd+Shift+E
Document Package	**Ctrl+Shift+D**	**Cmd+Shift+D**

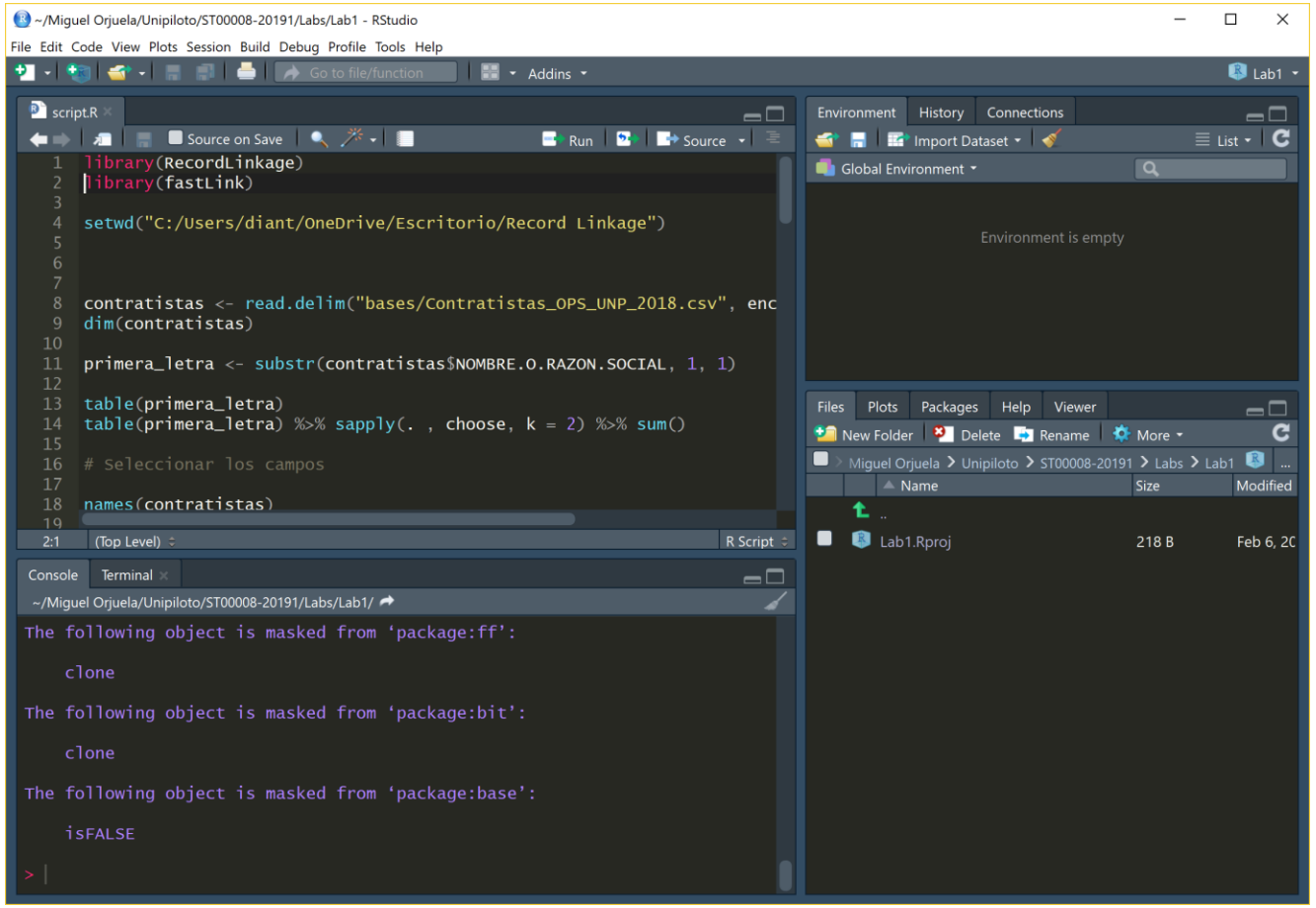
8 DOCUMENTS AND APPS

	Windows/Linux	Mac
Preview HTML (Markdown, etc.)	Ctrl+Shift+K	Cmd+Shift+K
Knit Document (knitr)	**Ctrl+Shift+K**	**Cmd+Shift+K**
Compile Notebook	Ctrl+Shift+K	Cmd+Shift+K
Compile PDF (TeX and Sweave)	Ctrl+Shift+K	Cmd+Shift+K
Insert chunk (Sweave and Knitr)	Ctrl+Alt+I	Cmd+Option+I
Insert code section	Ctrl+Shift+R	Cmd+Shift+R
Re-run previous region	Ctrl+Shift+P	Cmd+Shift+P
Run current document	Ctrl+Alt+R	Cmd+Option+R
Run from start to current line	**Ctrl+Alt+B**	**Cmd+Option+B**
Run the current code section	**Ctrl+Alt+T**	**Cmd+Option+T**
Run previous Sweave/Rmd code	Ctrl+Alt+P	Cmd+Option+P
Run the current chunk	Ctrl+Alt+C	Cmd+Option+C
Run the next chunk	Ctrl+Alt+N	Cmd+Option+N
Sync Editor & PDF Preview	Ctrl+F8	Cmd+F8
Previous plot	Ctrl+Alt+F11	Cmd+Option+F11
Next plot	Ctrl+Alt+F12	Cmd+Option+F12
Show Keyboard Shortcuts	**Alt+Shift+K**	**Option+Shift+K**

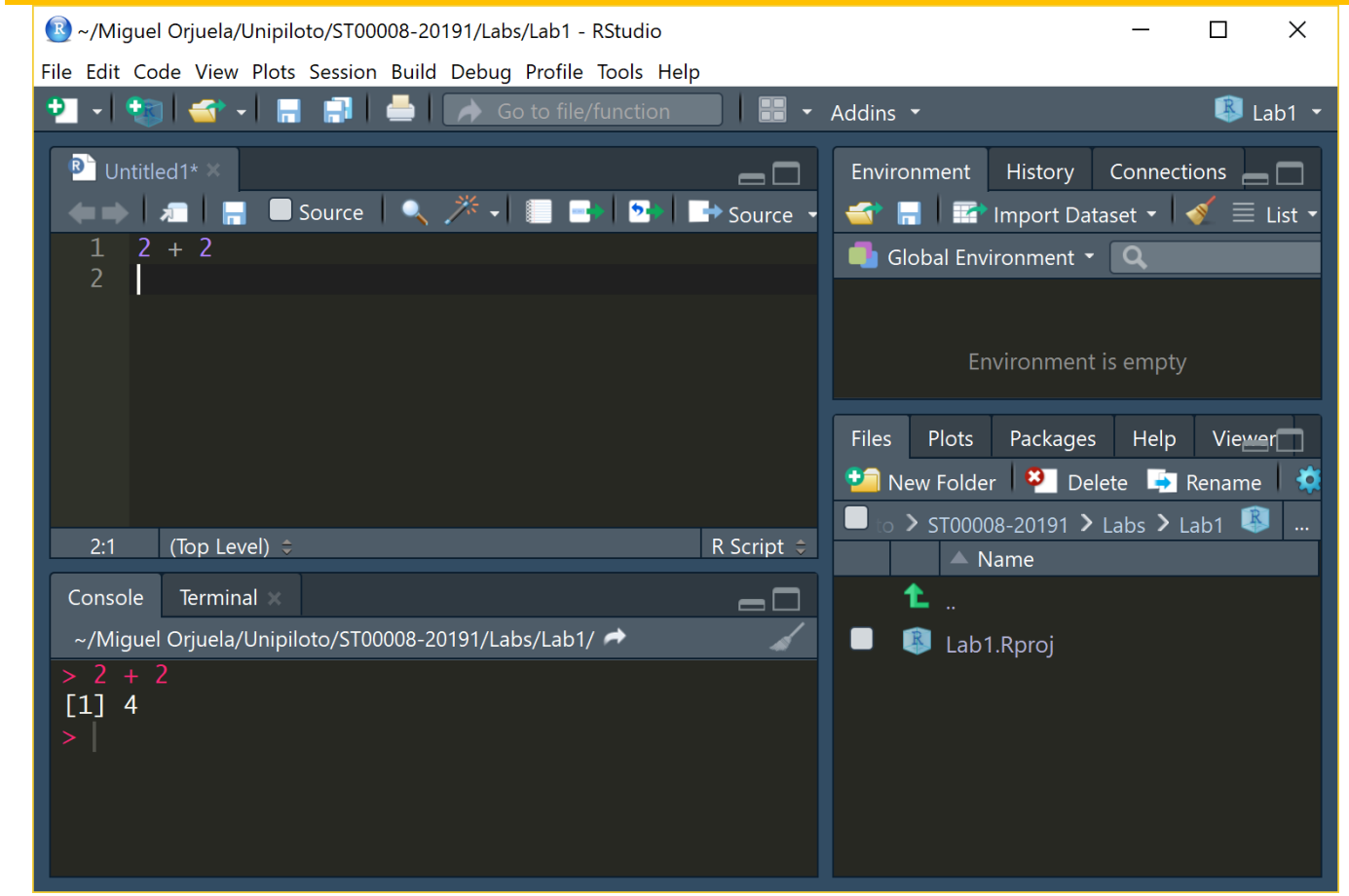
RStudio

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Interfaz de RStudio



Interfaz de RStudio



Paso 1: Escriba 2 + 2

Paso 2: Oprima Ctrl+Enter

Paso 3: Oprima Ctrl+L

Programación

1. Ingresando expresiones
2. Evaluación de expresiones
3. Objetos
4. Números
5. Atributos
6. Vectores
7. Objetos mixtos
8. Conversión explícita (Coercion)
9. Matrices
10. Listas
11. Factores
12. Valores faltantes/nulos (Missings)
13. Data frames
14. Nombres

Calculadora

```
2 + 2
sqrt(9)
log(10)
```

Ingresando expresiones

Operador de asignación **<-**

```
x <- 1
print(x)
x
msg <- "hola"
```

La gramática del lenguaje determina si una expresión está completa o no

Comentarios con #

```
# este es un comentario
```

Consejo: Alt + - inserta asignaciones

Evaluación de expresiones

Ingreso, evaluación, resultado impreso (autoprint)

```
x <- 5 # nada se imprime
x # auto-printing
print(x) # print explícito
```

```
y <- 11:40
y
```

¿Qué indica [1]?

[1] indica que x es vector y 6 es el primer elemento

Objetos

Clases básicas o **atómicas** de objetos

- character
- numeric (números reales)
- integer
- complex
- logical (true/false)

Vectores y listas

- vector – objetos de la misma clase
- list – objetos de diferente clase

Números

Números en R se tratan como objetos numéricos (reales de doble precision)

```
x <- 1
print(x)
x
msg <- "hola"
```

Si queremos hacer enteros le ponemos el surfijo L

```
y <- 1L
```

Verificar con commandos class, typeof, str

```
class(x)
typeof(x)
str(x)
```

Infinito **Inf**

Not a number **NaN**

```
1/0
log(0)
0/0
```

Atributos

Los objetos de R tienen metadata, muy útil para describir el objeto

Algunos atributos son:

- names, dimnames
- dimensions
- class
- length
- Otros definidos por el usuario

```r
x <- 1
print(x)
x
attributes(x)
```

Los atributos se consultan con **attributes()**. Si no tiene retorna **null**

Vectores

Función **c()** concatena objetos y hace vectores

```
x <- c(0.5,0.6) # numérico
x <- c(TRUE, FALSE) # lógico
x <- c(T, F) # lógico
x <- c("a", "b", "c") # caracteres
x <- 9:29 # enteros
x <- c(1+0i,2+4i) # complejos
```

Se puede usar la función **vector()** para inicializar vectores

```
x <- vector(mode = "numeric", length = 10)
```

Compruebe el tipo de x usando las funciones **class()**, **typeof()** y **str()**

Vectores

Creando vectores con objetos de diferente tipo

```
y <- c(1.7, "a") # caracteres
y <- c(TRUE, 2) # numericos
y <- c("a", TRUE) # caracter
```

Cuando esto sucede, R realiza una conversión implícita de los objetos

Vectores

Seleccionar elementos de los vectores

Ejemplo

```r
x <- seq(1, 10, by = 0.5)
```

- Por posición

```r
x[1]
x[4]
x[-4]
x[2:4]
```

- Por valor

```r
x[x == 10]
x[x > 3]
x[x %in% c(1,2,5)]
```

Conversión explícita

Funciones para hacer casting entre tipos de variables

```
x <- 0:6
class(x)
as.numeric(x)
as.logical(x)
as.character(x)
```

¿Qué pasa cuando no hay forma razonable de transformar los datos?

```
x <- c("a", "b", "c")
as.numeric(x)
as.logical(x)
as.complex(x)
```

Matrices

Son vectores con un atributo de dimension. La dimension es un vector de enteros de longitud 2 (num filas, num columnas)

```
m <- matrix(nrow = 2, ncol = 3)
m
dim(m)
attributes(m)
```

Las matrices son orientadas por columnas

```
m <- matrix(1:6, nrow = 2, ncol = 3)
m
```

Matrices

Se pueden crear de un vector agregandole dimension

```
m <- 1:10
m
dim(m) <- c(2,5)
m
```

También se crean pegando vectores filas o vectores columnas

```
x <- 1:3
y <- 10:12
cbind(x,y)
rbind(x,y)
```

Matrices

Seleccionar elementos particulares de una matriz

```
m <- 1:10
m
dim(m) <- c(2,5)
m
```

- Fila `m[2, ]`
- Columna `m[ ,1]`
- Elemento `m[2,3]`

Listas

Vectores con elementos de diferentes clases

```
x <- list(1, "a", TRUE, 1+4i)
x
```

```
x <- vector(mode = "list", length = 5)
x
```

Factores

Representan datos categóricos. Pueden ser desordenados u ordenados

```
x <- factor(c("yes","yes","no","yes","no"))
x
table(x)
```

¿Cómo están representados realmente?

```
unclass(x)
```

Factores

Agregando orden a las categorías

```
x <- factor(c("yes","yes","no","yes","no"))
x

x <- factor(c("yes","yes","no","yes","no"),
levels = c("yes","no"))
x
```

Valores faltantes/nulos (missings)

Detectar valores nulos y no numéricos

Un NaN es también un NA, pero un NA no es NaN

```
x <- c(1, 2, NA, 10, 3)
is.na(x)
is.nan(x)
```

```
x <- c(1, 2, NaN, NA, 4)
is.na(x)
is.nan(x)
```

Data frames

Tipo especial de matriz que puede guardar diferentes tipos de elementos en cada columna y todas las columnas tienen el mismo tamaño

```
x <- data.frame(foo = 1:4, bar = c(T, T, F, F))
x
nrow(x)
ncol(x)
```

```
data <- mtcars
class(data)
```

Names

Metadata que hace que los objetos se autodescriban

```
x <- 1:3
names(x)
x
names(x) <- c("New York", "Seattle", "Los Angeles")
x
```

```
x <- list("Los Angeles" = 1, Boston = 2, London = 3)
names(x)
x
```

Instrucciones condicionales

```
x <- runif(1,0,10)
```

```
if(<condition>) {
## hacer algo
} else {
## hacer algo diferente
}
```

```
if(<condition>) {
## hacer algo
} else {
## hacer algo diferente
}
```

```
if(<condition1>) {
## hacer algo
} else if(<condition2>) {
## hacer algo diferente
} else {
## hacer algo diferente
}
```

Ciclos

En R, los ciclos `for` toman una variable iteradora y le asignan valores sucesivos de una secuencia o vector.

Los ciclos `for` se usan más comúnmente para iterar sobre los elementos de un objeto (lista, vector, etc.)

```
for(i in 1:10) {
  print(i)
}
```

Ciclos

Los ciclos `while` inician verificando una condición, si se cumple, su bloque de código se ejecuta. Una vez ejecutado, se vuelve a verificar, y así sucesivamente hasta que la condición evalúa a falso.

```
count <- 0

while(count < 10) {
    print(count)
    count <- count + 1
}
```

Funciones

```
saludar <- function()
{
  print("hola")
}
saludar()
```

Sin retorno

```
sumar_dos <-
function(numero) {
  return(numero + 2)
}
sumar_dos(4)
```

Con retorno

Gracias por tu asistencia y participación ☺

Contacto

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Links de interés

- *https://bookdown.org/rdpeng/rprogdatascience/*
- *https://www.rstudio.com/resources/cheatsheets/*